



Royal Mail Group

Royal Mail Order V1 REST API

Technical User Guide

API Specification

This API specification details the requirements for integrating with the Royal Mail Order V1 REST API. It specifically covers how Order V1 REST API can be used by business customers to handle Parcel Collect Order booking and provides the technical information to build this integration.

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Version 0.4

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1 Document Control

1.1 Terms and Abbreviations

Term	Meaning
API	Application Program Interface
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol over TLS
IP	Internet Protocol
REST	Representational State Transfer
Swagger	Specification for defining RESTful web services
TLS	Transport Layer Security
URL	Uniform Resource Locator
XML	Extensible Markup Language

Table 1 – Terms and Abbreviations

1.2 Version History

Version	Date	Author	Notes
0.1	24/08/2020	RMG	Initial draft.
0.2	15/09/2020	RMG	Updated review comments from Bindhu
0.3	11/03/2022	RMG	Added New Create Label Operation
0.4	30/03/2022	RMG	Added guidance table for populating request data fields
0.5	17/10/2022	RMG	Booking flow diagram and allowed value list for “itemServicename”

Table 2 – Document Version History

2 Overview

The Royal Mail Order API exposes a REST service that allows third parties to create, update, retrieve or cancel a doorstep collection booking.

There are no costs to customers for using the Royal Mail Order API V1 services, however customers’ development costs must be covered by the customer developing the solution. Royal Mail will not accept any responsibility for this development, implementation and testing costs.

Customers should address initial enquiries regarding development of systems for these purposes to their account handler.

3 Purpose

This document is to provide Royal Mail customers with guidelines and detailed specifications for integrating with the Order V1 RESTful web service.

The document details:

- The specification for the web service interface is for providing consumers the facility to book and manage collections with Royal Mail.
- This API exposes the following capability:
 - ☐ Create a collection Order.
 - ☐ Retrieve a collection order to get the detailed information about existing collection.
 - ☐ Amend/Update Collection Order Details.
 - ☐ Cancel a collection Order.
- Description of errors the API can return.
- Non-functional characteristics of the API including response times, service availability and security considerations.

This document is primarily intended to be read by developers and other technical roles involved with integrating customer systems with the Order V1 API. This document should be read in conjunction with the following artefacts which are available from the 'Royal Mail Order V1 (REST)' page on the [Royal Mail API \(Developer\) Portal](#):

- Order V1 REST API Swagger Definition (Click on 'Open API' link on this page to download)

4 Introduction to Order V1 API

4.1 Overview

The Royal Mail Order API exposes a REST service that provides third parties the facility to create, update, retrieve or cancel collections with Royal Mail. The API validates and records the partner ID for OLP consumers and account number for third party retailers for invoicing purposes.

Note: Third party retailers who are not subscribed for Parcel Collect will not be able to use this service.

Also, there is a dependency on Address Matching Services based on Royal Mail PAF database for obtaining Delivery Point Suffix (DPS) for the collection address, as well as on Royal Mail API for obtaining a collection slot availability token. Refer to the Appendix for specific details.

4.2 Interface Components

Please see Figure 1 below for a graphical representation of the interface between Royal Mail and customers for Order V1 API. This document covers what information is to be exchanged, how this information is structured and the means by which it is transferred.

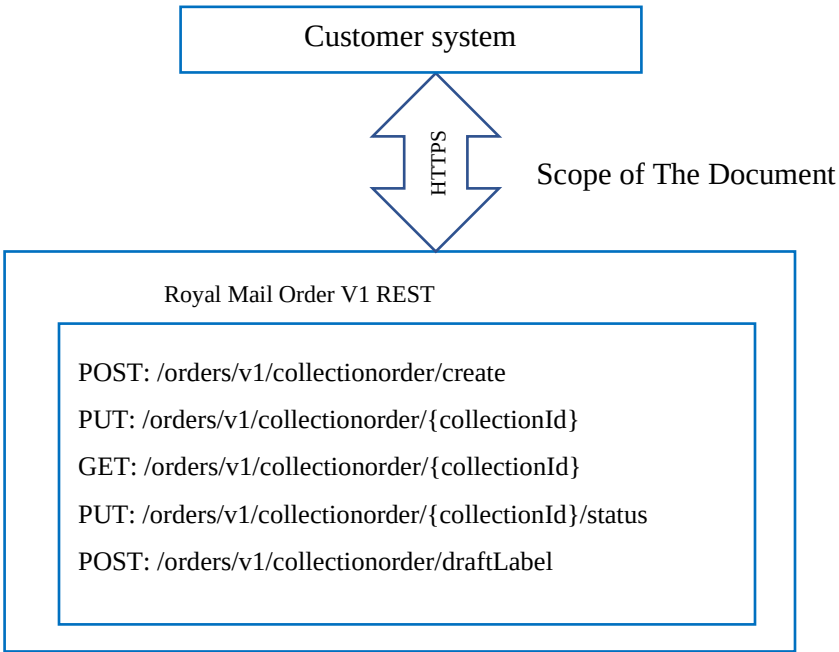


Figure 1 – Royal mail Order V1 API

5 Integrating with Order V1 API

5.1 On-boarding Process

The high-level process associated with integrating Order V1 API is represented and described in the diagram below.

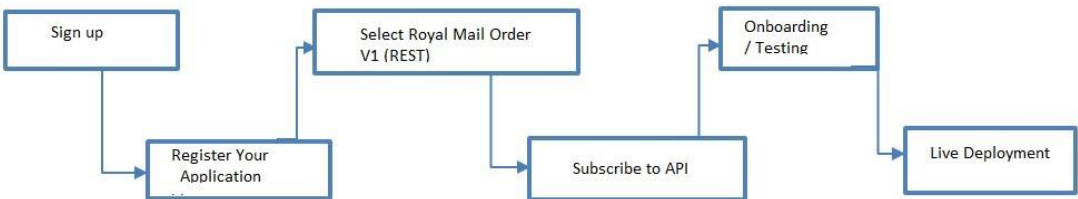


Figure 2 – Process for Integrating with the API

Access to the service is managed through RMG’s API Management system.

New users of the system will need to:

1. Sign up for an account and accept the terms and conditions on the [Royal Mail API \(Developer\) Portal](#).
2. Register the 'application' which will be calling the API. When the application is registered, it will be assigned a unique system-generated Client ID and Client Secret which is needed to securely access the API. It is important that these credentials are noted and securely stored.
3. Request to subscribe to the API. This will result in an e-mail being automatically generated and sent to the Royal Mail Customer Solutions team.
4. Once approved, testing can be performed against the API at controlled transaction rate.
5. Once all required testing/integration has completed, access to the Live API will be provided at a mutually agreed date/time.

Existing users who already have an account with Royal Mail's API Management system will need to perform step 2 onwards if the application accessing the API is different to any currently registered applications. If the application accessing the API is already registered, existing customers will need to perform step 3 onwards.

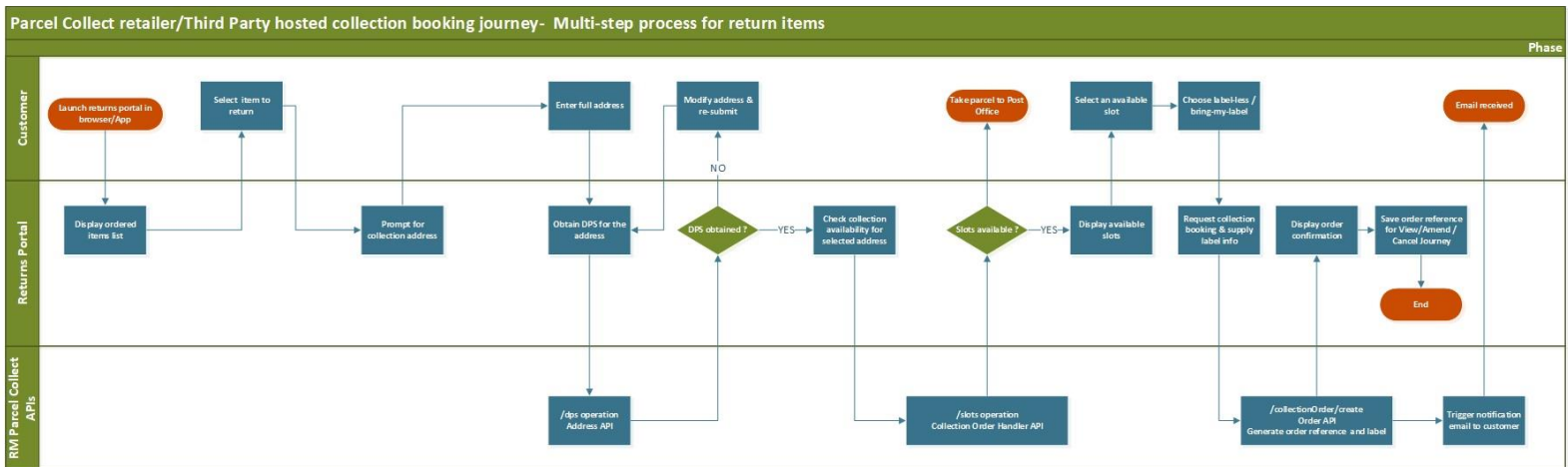
You will be provided with a contact in Royal Mail who will take you through the on-boarding process.

5.2 Terms & Conditions

You must accept the Royal Mail Terms and Conditions when creating your customer account. These cover the ways in which the service may be used and any integration activities must abide by these.

Of particular note to developers:

- The API is available 24x7 although must not be used for performance testing.
- The API imposes a cap on the number of transactions per second for each customer. Excessive volumes of traffic over the configured throttling limit will result in transactions being rejected.
- Royal Mail expects customers to use the service in a responsible way; this includes refraining from continuously polling the API for updates.
- Regular checks will be performed by Royal Mail to ensure that customers are using the API in a responsible way.
- Some of the data fields in the API request messages might have been specified as optional fields in the swagger document (YAML file), but could be mandatory based on certain conditions. Refer to **"Guide to populating Request data fields"** in the Appendix section of this document. The diagram below shows the steps normally involved in a collection booking journey, including the sequence in which Royal Mail Parcel collect APIs have to be called.



5.3 API Access

Access to the Order V1 API is via the following URL:

POST: <https://api.royalmail.net/orders/v1/collectionOrder/create>

GET: <https://api.royalmail.net/orders/v1/collectionOrder/{collectionId}>

PUT: <https://api.royalmail.net/orders/v1/collectionOrder/{collectionId}>

PUT: <https://api.royalmail.net/orders/v1/collectionOrder/{collectionId}/status>

POST: <https://api.royalmail.net/orders/v1/collectionOrder/draftLabel>

The API only supports HTTP GET, POST, PUT requests against above paths. Please note:

- Response formats supported: application/json
- See section 6 for all operations and parameters which are supported by this API.
- If anything is appended to the path and which is unsupported (e.g. /orders/something), then a HTTP 404 Not Found response will be returned.
- The Client must be provided in the HTTP header of all API requests otherwise access to the API will be rejected and a HTTP 401 (Unauthorised) will be returned. The Client ID is obtained by registering an application on the [Royal Mail API \(Developer\) Portal](#).

5.4 API Versioning

Royal Mail is continuously working to improve its technology, and as part of this process updates to the services provided may on occasion necessitate a new API version. Royal

Mail will look to maintain three versions of the API; as new versions are introduced; previous versions move down the stack until they are ultimately removed completely:

- Latest version
- Previous version
- Deprecated version

Customers will always be encouraged to integrate against the latest version as this will give them the longest stable period without the need to change, but if they have already begun integration activities when a new version is released then they will be able to integrate against the previous version. Customers should not integrate against the deprecated version.

6 Order Services

6.1 Business Services

The Order V1 API is a service offered to handle Parcel Collect order booking for third party retailers by calling Order API on Store Feeder. This provides the ability to create, amend, retrieve and amend order. The table below provides an overview of the business services that are supported by this interface.

Business Service	API Operations	Description	Technology	Conversation Style
Create	POST: /orders/v1/collectionOrder/create	Creates a collection order with a unique collection id	JSON over HTTPS (REST)	Synchronous Request / Response
Get	GET: /orders/v1/collectionOrder/{collectionId}	Retrieves a collection order with given collectionId	JSON over HTTPS (REST)	Synchronous Request / Response
Update	PUT: /orders/v1/collectionOrder/{collectionId}	Updates a collection order with given collectionId	JSON over HTTPS (REST)	Synchronous Request / Response
Cancel	PUT: /orders/v1/collectionOrder/{collectionId}/status	Cancel an existing collection Order	JSON over HTTPS (REST)	Synchronous Request / Response

Create Label Item	POST: /orders/v1/collectionOrder/draftLabel	Create a draft collection label item.	JSON over HTTPS (REST)	Synchronous Request / Response
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Table 3 – Business Services

6.2 HTTP Header Information

6.2.1 Description

The purpose of the HTTP header is to support security and logging functionality within the Royal Mail systems and it is mandatory that it is provided in the request message.

6.2.2 Request Headers

All service requests to this API will be authorised in accordance with the Client ID and Secret passed in the HTTP headers. Please see table below for the elements which need to be populated in the HTTP header.

Parameter	Optional	Description
X-IBM-Client-Id	No	Similar to a client username. Required to access the API.
X-IBM-ClientSecret	No	Similar to a client password. Required to access the API.
X-RMG-Date-Time	No	This should be populated with the date time in ISO 8601 format
X-RMG-Language	Yes	Optional default english
Accept	Yes	Content-Types that are acceptable for the response. Eg. application/json

Table 4 – HTTP Header Information in the API Request

6.2.3 Header Request Example Data

Example Request Data for the HTTP Header:

Parameter	Value
X-IBM-Client-Id	f0e***51-2041-4df2-b31d
X-IBM-Client-Secret	**0e***51-2041-4df2-b31d
X-RMG-Date-Time	2016-10-20T10:04:00+01:00
X-RMG-Language	en
Accept	Application/json

Table 5 – Example HTTP Header Information for API Request

6.2.4 Response Headers

Please see table below for the elements which are populated in the HTTP response header.

Parameter	Optional	Description
HTTP/1.1	No	HTTP status code and reason phrase
X-BacksideTransport	No	OK OK on success or FAIL, FAIL on failure
Connection	No	Controls whether the network connection stays open after the current transaction finishes.
Content-Length	No	Indicates the content length
Content-Type	No	Indicates the media type of the resource.
Date	No	Contains the date and time at which the message was originated.
X-Global-Transaction-ID	No	Global transaction id
Access-Control-Expose-Headers	No	Indicates which headers can be exposed as part of the response by listing their names.
Access-Control-Allow-Origin	No	Indicates whether the response can be shared.
Access-Control-Allow-Methods	No	This response header specifies the method or methods allowed when accessing the resource.
X-RateLimitLimit	No	Request limit per day / per 5 minutes
X-RateLimitRemaining	No	The number of requests left for the time window

Table 6 – HTTP Header Information in the API Response

6.2.5 Header Response Example Data

Example Response Data for the HTTP Header:

Parameter	Value
HTTP/1.1	200 Ok
X-Backside-Transport	OK OK
Connection	Keep-Alive
Content-Length	4566
Content-Type	application/json
Date	14 Feb 2020 14:33:16 GMT
X-Global-Transaction-ID	5s****fd8*****afacae13d469

Access-Control-Expose-Headers	APIm-Debug-Trans-Id
Access-Control-Allow-Origin	*.royalmail.com
Access-Control-Allow-Methods	POST
X-RateLimit-Limit	rate-limit-1,220;
X-RateLimit-Remaining	name=rate-limit-1,219;

Table 7 – Example HTTP Header Information for API Response

6.3 Create Operation

6.3.1 Description

The behaviour of this operation is to create a new collection Order.

6.3.2 Request Message

URL Path POST: <https://api.royalmail.net/orders/v1/collectionOrder/create>

Please refer to section 6.2 for the request and response header information.

Parameter Name	Optional	Type	Description
Collection	No	Object	Request body to create a new collection order. For more details check the example below

Table 8 – Request message to Create operation

6.3.3 Response Message

A successful business response will be returned as a standard HTTP response code of 201 (Created).

Parameter Name	Optional	Type	Description
responseBody	No	Object	Response for create collection order. For more details check the example below

Table 9 – Response message for Create operation

6.3.4 Create Operation Example Data

Full JSON example responses are provided on the [Royal Mail API \(Developer\) Portal](#).

Example Request Data

POST https://api.royalmail.net/orders/v1/collectionOrder/create HTTP/1.1
Accept-Encoding: gzip,deflate
Content-Type: application/json
X-RMG-Date-Time: 2016-10-20T10:04:00+01:00
X-IBM-Client-ID: 00*****1-5**9-4**1-9**4-6*****0
X-IBM-Client-Secret: hU2goA3rK8bY0*****oW
Content-Length: 916
Host: api.royalmail.net
Connection: Keep-Alive
User-Agent: Apache-HttpClient/4.1.1 (java 1.5)

```
{
  "timeslotReservationId": " 16566688308323200147",
  "senderDetails": {
    "senderName": "John Solly",
    "senderEmail": "john.solly@royalmail.com"
  },
  "accountDetails": {
    "retailerAccountNumber": "123456789"
  },
  "address": {
    "addressLine1": " Flat 3, Kenyatta Court ",
    "addressLine2": ", 21 Harrington Street ",
    "postTown": "Derby",
    "postcode": "DE238EB",
    "DPS": "2N",
    "County": "Derbyshire"
  },
  "safePlaceDetails": {
    "locationText": "behind the bin",
    "locationCode": "BHDBIN"
  },
  "animalHazardDetails": "Dogs - not dangerous",
  "collectionDate": "2022-06-30",
  "items": [
    {
      "itemBarcodeId": "WL902730228GB",
      "weightInGrams": 100,
      "itemServiceName": "Tracked Returns 24 (T24) Enhanced",
      "itemStatus": "AwaitingCollection",
      "dimensions": {
        "height": 45,
        "width": 35,
        "depth": 16
      },
      "itemPrice": 0,
      "itemType": "NoLabel",
      "itemProductCode": "ABC123-XY",
      "labelInfo": {
        "1DTrackingNumber": "HZ000120140GB",
        "2DUniqueIdentifier": "1B0572355000C44171776",
        "postByDate": "2022-05-19",
```

```

        "RMService": "Tracked Returns 48 (T48) Enhanced",
        "pricePaid": 1.99,
        "referenceNumber": "AAA-BBB-11-123-C",
        "referenceText": "Online order number",
        "weightInGrams": 20000,
        "itemFormat": "Letter",
        "recipientAddress": {
            "companyName": "R",
            "addressLine1": "The White House 162 North Lane",
            "postTown": "Aldershot",
            "county": "string",
            "postcode": "GU124QR",
            "DPS": "R1",
            "countryCode": "GBR"
        },
        "senderAddress": {
            "companyName": "A B C",
            "addressLine1": "57 Mill Hill Lane",
            "postTown": "Derby",
            "county": "Derbyshire",
            "postcode": "DE238EB",
            "DPS": "2N ",
            "countryCode": "GBR"
        }
    },
    "suppressRmNotifications": false
}

```

Example Response Data

```

HTTP/1.1 201 Created
X-Backside-Transport: OK OK
Connection: Keep-Alive
Content-Length: 1232
Content-Type: application/json
Date: Mon, 24 Aug 2020 06:36:31 GMT
X-Global-Transaction-ID: 5521cfd85f435fef1345a631
Access-Control-Expose-Headers: APIm-Debug-Trans-Id, X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset, X-Global-Transaction-ID
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: POST
X-RateLimit-Limit: name=rate-limit-1,120;
X-RateLimit-Remaining: name=rate-limit-1,119;
Set-Cookie: DPJSESSIONID=PBC5YS:464726495; Path=/; Domain=.royalmail.net

{"collectionOrderId":"52417","status":"Order created successfully"}

```

6.4 Get Operation

6.4.1 Description

The behaviour of this operation is to get collection details for a given collection order id.

6.4.2 Request Message

URL Path: GET: <https://api.royalmail.net/orders/v1/collectionOrder/{collectionId}>

Please refer to section 6.2 for the request and response header information.

Parameter Name	Optional	Type	Description
collectionId	No	String	Unique collection identifier of the collection order being queried.

Table 8 – Request message for GET operation

6.4.3 Response Message

A successful business response will be returned as a standard HTTP response code of 200 (Ok).

Parameter Name	Optional	Type	Description
responseBody	No	Object	Response for Get collection order. For more details check the example below

Table 9 – Response message for GET operation

6.4.4 Get Operation Example Data

Full JSON example responses are provided on the [Royal Mail API \(Developer\) Portal](#).

Example Request Data

```
GET https://api.royalmail.net/orders/v1/collectionOrder/CC-W**7-00*****2
HTTP/1.1
Accept-Encoding: gzip,deflate
X-RMG-Date-Time: 2016-10-20T10:04:00+01:00
X-IBM-Client-ID: 00****a1-5**9-4**1-9**4-6fc3*****500
X-IBM-Client-Secret: hU2gW8hJ1*****8bY0sX2qA*****oW
Host: api.royalmail.net
Connection: Keep-Alive
User-Agent: Apache-HttpClient/4.1.1 (java 1.5)
```

Example Response Data

GET response:
HTTP/1.1 200 OK
X-Backside-Transport: OK OK
Connection: Keep-Alive
Content-Length: 4556
Content-Type: application/json
Date: Mon, 24 Aug 2020 06:29:06 GMT
X-Global-Transaction-ID: 5521cfd85f435e310fe65945
Access-Control-Expose-Headers: APIm-Debug-Trans-Id, X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset, X-Global-Transaction-ID
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET
X-RateLimit-Limit: name=rate-limit-1,120;
X-RateLimit-Remaining: name=rate-limit-1,119;
{
 "collectionOrderId": "ABC123",
 "timeslotReservationId": "52353",
 "senderDetails": {
 "senderName": "John Solly",
 "senderEmail": "john.solly@royalmail.com"
 },
 "accountDetails": {
 "partnerId": "WL902730228GB",
 "retailerAccountNumber": "123456789"
 },
 "address": {
 "addressLine1": "Southend Road",
 "addressLine2": "EAST HAM",
 "addressLine3": "addressLine3",
 "postTown": "London",
 "postcode": "E62AL",
 "DPS": "BD1",
 "County": "Norfolk"
 },
 "safePlaceDetails": {
 "locationText": "behind the bin",
 "locationCode": "BHDBIN"
 },
 "animalHazardDetails": "Dogs - not dangerous",
 "collectionStatus": "created",
 "collectionDateTime": "2020-05-07T05:54:21.100Z",
 "items": [
 {
 "itemBarcodeId": "WL902730228GB",
 "weightInGrams": 100,
 "itemServiceName": "Tracked Returns 24 (T24) Enhanced",
 "itemStatus": "AwaitingCollection",
 "dimensions": {
 "height": 45,
 "width": 35,
 "depth": 16
 },
 "itemPrice": 0,
 "eventStatusCode": "strin",
 "itemType": "Standard",
 "customerCode": "string",
 "itemProductCode": "strin",


```

"labelInfo": {
  "1DTrackingNumber": "string",
  "2DUniqueIdentifier": "string",
  "postByDate": "2022-03-25",
  "RMService": "string",
  "pricePaid": 0,
  "referenceText": "string",
  "itemFormat": "Letter",
  "recipientAddress": {
    "companyName": "string",
    "addressLine1": "string",
    "addressLine2": "string",
    "addressLine3": "string",
    "postTown": "string",
    "county": "string",
    "postcode": "string",
    "DPS": "9Z",
    "countryCode": "GBR"
  },
  "senderAddress": {
    "companyName": "string",
    "addressLine1": "string",
    "addressLine2": "string",
    "addressLine3": "string",
    "postTown": "string",
    "county": "string",
    "postcode": "string",
    "DPS": "9Z",
    "countryCode": "GBR"
  },
  "customsInfo": {
    "currencyCode": "str",
    "totalCost": 0,
    "shipmentSubtotal": 0,
    "shippingCostCharged": 0,
    "customsDutyCost": 0,
    "IOSSNumber": "string",
    "AIRNumber": "string",
    "customsDeclarationCategory": "none",
    "contents": [
      {
        "description": "string",
        "customsCode": "string",
        "quantity": 0,
        "unitValue": 0,
        "originCountryCode": "str"
      }
    ]
  }
},
"createOn": "2020-05-04T13:12:30.873Z",
"lastModifiedOn": "2020-05-04T13:12:30.873Z",
"suppressRmNotifications": true
}

```

6.5 Update Operation

6.5.1 Description

The behaviour of this operation is to Amend/update collection order details for a given collectionId. Only collection date and collection address can be amended, but not the count / details of the items in the collection order. The cut-off time for amending the order is till midnight before the day of the collection.

6.5.2 Request Message

URL Path: PUT: <https://api.royalmail.net/orders/v1/collectionOrder/{collectionId}>.

Please refer to section 6.2 for the request and response header information.

Parameter Name	Optional	Type	Description
collectionId	No	String	Unique collection identifier of the collection order being queried.
Collection	No	Object	Request to Update an existing collection order details. For more details check the example below

Table 10 – Request message for Update operation

6.5.3 Response Message

A successful business response will be returned as a standard HTTP response code of 200 (Ok).

Parameter Name	Optional	Type	Description
responseBody	No	Object	Response Object for Update collection order details. For more details check the example below

Table 11 – Response message for Update operation

6.5.4 Update Operation Example Data

Full JSON example responses are provided on the [Royal Mail API \(Developer\) Portal](#).

Example Request Data

```

PUT https://api.royalmail.net/orders/v1/collectionOrder/CC-W**7-00*****2 HTTP/1.1
Accept-Encoding: gzip,deflate
Content-Type: application/json
X_RMG-Language: en
Accept: application/json
X-RMG-Date-Time: 2016-10-20T10:04:00+01:00
X-IBM-Client-Id: 331*****5-8**8-4**2-a**7-e24f*****8
X-IBM-Client-Secret: vM6f*****fG2gX4rQ2*****3uU8eD6pW3u*****Y
Content-Length: 539
Host: api.royalmail.net
Connection: Keep-Alive
User-Agent: Apache-HttpClient/4.1.1 (java 1.5)
{
  "timeslotReservationId": "52353",
  "senderDetails": {
    "senderName": "John Solly",
    "senderEmail": "john.solly@royalmail.com"  },
  "address": {
    "addressLine1": "Southend Road",
    "addressLine2": "EAST HAM",
    "addressLine3": "addressLine3",
    "postTown": "London",
    "postcode": "E62AL",
    "DPS": "BD1",
    "County": "Norfolk"
  },
  "safePlaceDetails": {
    "locationText": "behind the bin",
    "locationCode": "BHDBIN"
  },
  "animalHazardDetails": "Dogs - not dangerous",
  "collectionDate": "2017-11-16" }

```

Example Response Data

```

HTTP/1.1 200 OK
X-Backside-Transport: OK OK
Connection: Keep-Alive
Content-Length:1343
Content-Type: application/json
Date: Mon, 24 Aug 2020 06:52:38 GMT
X-Global-Transaction-ID: 5521cfd85f4363b60fef59e5
Access-Control-Expose-Headers: APIm-Debug-Trans-Id, X-RateLimit-Limit, X-
RateLimit-Remaining, X-RateLimit-Reset, X-Global-Transaction-ID
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: PUT
X-RateLimit-Limit: name=rate-limit-1,120;
X-RateLimit-Remaining: name=rate-limit-1,119;
{"collectionOrderId":"ABC201","status":"Order updated successfully"}

```

6.6 Cancel Operation

6.6.1 Description

The behaviour of this operation is to cancel a collection order id by giving an existing collection Id and status as cancelled. The cut-off time for cancelling the order is till midnight before the day of the collection.

6.6.2 Request Message

URL Path: PUT: <https://api.royalmail.net/orders/v1/collectionOrder/{collectionId}/status>

Please refer to section 6.2 for the request and response header information.

Parameter Name	Optional	Type	Description
collectionId	No	String	Unique collection identifier of the collection order being queried.
Collection Status Request	No	Object	Request to update the status of an existing collection order to cancelled. For more details check the example below

Table 10 – Request message for Cancel operation

6.6.3 Response Message

The body of the response message can contain a success response message or error message as described in the Error! Not a valid bookmark self-reference. section. A successful business response will be returned as a standard HTTP response code of 200 (Ok).

Parameter Name	Optional	Type	Description
responseBody	No	Object	Response object for cancel collection order. For more details check the example below

Table 11 – Response message for Cancel operation

6.6.4 Cancel Operation Example Data

Full JSON example responses are provided on the [Royal Mail API \(Developer\) Portal](#).

Example Request Data

```
PUT https://api.royalmail.net/orders/v1/collectionOrder/CC-W**7-00****2/status HTTP/1.1
Accept-Encoding: gzip,deflate
Content-Type: application/json
Accept: application/json
X-RMG-Date-Time: 2016-10-20T10:04:00+01:00
X-RMG-Language: en
X-IBM-Client-Id: 331****5-8**8-4**2-a**7-e24f*****8
X-IBM-Client-Secret: vM6f*****fG2gX4rQ2*****3uU8eD6pW3u*****Y
{
  "status": "Cancelled",
}
```

Example Response Data

```
HTTP/1.1 200 OK
X-Backside-Transport: OK OK
Connection: Keep-Alive
Content-Length: 1323
Content-Type: application/json
Date: Mon, 24 Aug 2020 06:52:38 GMT
X-Global-Transaction-ID: 5521cfd85f4363b60fef59e5
Access-Control-Expose-Headers: API-Debug-Trans-Id, X-RateLimit-Limit, X-
RateLimit-Remaining, X-RateLimit-Reset, X-Global-Transaction-ID
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: PUT
X-RateLimit-Limit: name=rate-limit-1,120;
X-RateLimit-Remaining: name=rate-limit-1,119;
{"collectionOrderId":"ABC201","status":"Order cancelled successfully"}
```

6.7 Create Label Item

6.7.1 Description

The behaviour of this operation is to create a draft collection label item.

6.7.2 Request Message

URL Path POST: <https://api.royalmail.net/orders/v1/collectionOrder/draftLabel>

Please refer to section 6.2 for the request and response header information.

Parameter Name	Optional	Type	Description
labelInfo	No	Object	To create a draft collection label item.

Table 12 – Request message for Draft Label

6.7.3 Response Message

The body of the response message can contain a success response message or error message as described in the Error! Not a valid bookmark self-reference. section. A successful business response will be returned as a standard HTTP response code of 200 (Ok).

Parameter Name	Optional	Type	Description
responseBody	No	Object	Returns the draftId. For more details check the example below.

Table 11 – Response message for Draft Label

6.7.4 Create Label Item Example Data

Full JSON example responses are provided on the [Royal Mail API \(Developer\) Portal](#).

Example Request Data

```
POST https://api.royalmail.net/orders/v1/collectionOrder/draftLabel HTTP/1.1
Accept-Encoding: gzip,deflate
Content-Type: application/json
X-RMG-Date-Time: 2016-10-20T10:04:00+01:00
X-IBM-Client-ID: 00*****1-5**9-4**1-9**4-6*****0
X-IBM-Client-Secret: hU2goA3rK8bY0*****oW
Content-Length: 916
Host: api.royalmail.net
Connection: Keep-Alive
User-Agent: Apache-HttpClient/4.1.1 (java 1.5)
{
  "1DTrackingNumber": "string",
  "2DUniqueIdentifier": "string",
  "referenceNumber": "string",
  "displayReferenceAsBarcode": true,
  "referenceText": "string",
  "weightInGrams": 20000,
  "format": "Letter",
  "recipientAddress": {
    "companyName": "string",
    "addressLine1": "string",
    "addressLine2": "string",
    "addressLine3": "string",
    "postTown": "string",
    "county": "string",
    "postcode": "string",
    "DPS": "9Z",
    "countryCode": "GBR"
  }
}
```

Example Response Data

```
HTTP/1.1 200 OK
X-Backside-Transport: OK OK
Connection: Keep-Alive
Content-Length: 1323
Content-Type: application/json
Date: Mon, 24 Aug 2020 06:52:38 GMT
X-Global-Transaction-ID: 5521cfd85f4363b60fef59e5
Access-Control-Expose-Headers: API-Debug-Trans-Id, X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset, X-Global-Transaction-ID
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: PUT
X-RateLimit-Limit: name=rate-limit-1,120;
X-RateLimit-Remaining: name=rate-limit-1,119;

{"draftId": "string"}
```

7 Error Handling

7.1 Overview

There are two types of errors produced by the Order V1 API, namely:

- Business Errors (e.g. collectionId invalid etc.)
- Technical Errors / Exceptions (e.g. service unavailable etc.). There are two types of technical errors:
 1. API Gateway Errors. Returned when the API request is rejected by the API Gateway before any business processing takes place. E.g. rate limit exceeded.
 2. Application Errors. Returned when API requests are accepted for business processing but processing cannot be completed due to a technical error. E.g. internal server error.

Both sets of errors should be appropriately handled by your system and technical details of the error should not be displayed directly to consumers. Please refer to the tables below for the generic structure of all core error messages returned by Order V1 API.

The tables below show the fields for API Gateway and non-API Gateway type error responses.

API Gateway Error

Field	Optional	Type	Description
httpCode	No	String	HTTP error code
httpMessage	No	String	HTTP error code description
moreInformation	Yes	String	Information relating to the error condition

Table 12 – API Gateway Error Response Structure

Non-API Gateway Error (Business & Application)

Field	Optional	Type	Description
Errors	Yes	Array	Array containing error information
errors.code	Yes	String	Code associated with the error condition Yes
errors.description	Yes	String	Description of the error condition
Field	Optional	Type	Description
errors.cause	Yes	String	Cause of the error (if known)

errors.resolution	Yes	String	Description of the resolution and action required to correct the error
-------------------	-----	--------	--

Table 13 – Non-API Gateway Error Response Structure

The following sections describe the content of the business and technical errors returned by Order V1 API with their respective HTTP status code response.

7.2 Business Errors

All errors associated with processing the API requests are returned using the error structure defined in section 7.1.

For missing or empty mandatory field's scenario:

HTTP Status Code	code	description	cause	resolution
400	E1398	[Field] is required and must not be empty	Missing or empty mandatory field	Check mandatory field(s) and resubmit
400	E1704	The 'CollectionDateTime' parameter value cannot be in the past. CC-Correlation-Id:a2*****a-***d-4e**-*****d	Invalid request parameters	Please check and re-submit
400	E1705	Collection Item count should be {count} or less	Invalid request parameters	Please check and re-submit
400	E1706	Invalid created collection	Invalid request parameters	Please check and re-submit
403	E0102	Collection from your location being unavailable for the date selected	N/A	N/A
403	E0103	An error encountered whilst making your booking	N/A	N/A

404	E0101	The address provided being unrecognised	N/A	N/A
404	E1701	collection with CC-W**70*****2 not found	Details are not found for the requested parameters	Please try again later
404	E1702	collection with CC-W**70*****2 not found	Details are not found for the requested parameters	Please try again later
404	E1703	collection with CC-W**70*****2 not found	Details are not found for the requested parameters	Please try again later

Table 14 – API Business Errors

7.2.1 Example Data

Please see below for an example of a business error which is returned which results in an E1329 error being returned. Full JSON example responses are provided on the [Royal Mail API \(Developer\) Portal](#).

```

HTTP/1.1 400
Content-Type: application/json

{
  "statusCode": "400",
  "httpMessage": "Bad Request",
  "errors": [
    {
      "code": "E1398",
      "description": "[Field] is required and must not be empty",
      "cause": "Missing or empty mandatory field",
      "resolution": "Check mandatory field(s) and resubmit"
    }
  ]
}

```

7.3 Technical Errors / Exceptions

The following technical exceptions/error scenarios will be caught and handled as described in the tables below.

The tables below show API Gateway and Application technical errors. Only the most common API Gateway error descriptions are shown, with the HTTP status return code considered adequate for other cases and hence other response codes and descriptions are not listed here.

API Gateway Errors

HTTP Status Code	httpCode	httpMessage	moreInformation
401	401	Unauthorized	Invalid client id or secret
404	404	Not Found	API not found for requested URI
405	405	Method Not Allowed	The method is not allowed for the requested URL
429	429	Too Many Requests	The rate limit has been exceeded for the plan or operation being used.

Table 15 – API Gateway Technical Errors

Application Errors

HTTP Status Code	code	description	cause	resolution
500	E0000	Internal exception occurred	An internal error was identified while attempting to process your API request	Please try again later
503	E0001	Service unavailable	An internal error was identified while attempting to process your API request	Please try again later

Table 16 – Application Technical Errors

For all other technical issues please contact a Royal Mail Support representative by visiting the [Royal Mail API \(Developer\) Portal Support](#) pages.

7.3.1 Example Data

Please see below for examples of both API Gateway and NonGateway technical errors which are returned in the event that the rate limit for API calls has been exceeded or if an internal error is encountered whilst processing the request.

Full JSON example responses are provided on the [Royal Mail API \(Developer\) Portal](#).

API Gateway Error

```
HTTP/1.1 429
Content-Type: application/json

{
  "httpCode": "429",
```

```
"httpMessage": "Too Many Requests",  
"moreInformation": "The rate limit has been exceeded for the plan or operation being used."  
}
```

Non-API Gateway Error

```
HTTP/1.1 500  
Content-Type: application/json  
  
{  
  "statusCode": "500",  
  "httpMessage": "Internal server Error",  
  "errors": [  
    {  
      "code": "E0000",  
      "description": "Internal exception occurred",  
      "cause": "An internal error was identified while attempting to process your API request",  
      "resolution": "Please try again later"  
    }  
  ]  
}
```

8 Non-Functional Characteristics

8.1 Availability

8.1.1 Service Hours

The Order API is available 24 hours per day x 365 days per year.

8.1.2 Maintenance Windows

Royal Mail Online Services Terms and Conditions define the maintenance for this service.

8.1.3 Unavailability

In the unlikely event of the API being unavailable, customer systems should make provision to handle this appropriately.

If you experience issues with the availability of this API please contact a Royal Mail Support representative by visiting the [Royal Mail API \(Developer\) Portal Support](#) pages.

8.2 Performance

Royal Mail aims to respond to calls in less than 5 seconds on average when invoked from the edge of Royal Mail's UK data centre.

8.3 Security

The REST API will only accept requests and return responses over HTTPS. All service requests via the API Management solution will be authorised in accordance with the below security parameters passed in the HTTP headers. This will ensure that any external service requests are authorised and authenticated in line with RMG Security Policies and Standards.

1. Client Id and client secret

9 Frequently Asked Questions

Please see the [FAQ page](#) on the [Royal Mail API \(Developer\) Portal](#) for a general list of frequently asked questions with responses.

All FAQs specific to the API described in this document are listed below.

9.1 Latest versions of Swagger Definition

Question: Where can I find the latest version of the Order V1 API Swagger definition?

Answer: The latest version of the Swagger definition can be found on the 'Royal Mail Order V1 (REST)' page on the [Royal Mail API \(Developer\) Portal](#). The 'Open API' link can be used to download the swagger.

9.2 Sample Code

Question: Do Royal Mail provide any sample code for the Order V1 API to help accelerate my integration?

Answer: Yes, the Royal Mail API (Developer) Portal provides sample code for a number of programming languages including cURL, PHP, Ruby, Python, Java, Node, Go, Swift etc. Please navigate to the 'API Library' and select 'Royal Mail Order V1 REST API'. A variety of programming languages will be provided on the right hand-side of the page.

9.3 API Programming

Question: Can Royal Mail complete the API programming for me?

Answer: Royal Mail only provides user guides to enable an understanding of the API and therefore cannot complete any programming or system development for your business.

9.4 Business Account

Question: I don't have a business account with Royal Mail. Can I use the Order V1 API?

Answer: No – Order V1 API is only available to Royal Mail account holders.

9.5 Application Compatibility

Question: What Software Development Kits or tools have been proven to work with the Order V1 API?

Answer: The following applications are known to be compatible with the Royal Mail Order V1 API: SoapUI.

10 Appendix

a Guide to populating Request data fields

Tags	Description	Mandatory(Y/N)	Sample Values	Comments
Header				
X-IBM-Client-Id	Similar to a client username . Required to access the API	Y	Will be available after registering in the RMG API Developer portal.	The key pair created with registering the App on RM developer portal provides the client ID and Secret. The app needs to have a valid subscription to Order API in order to access the API successfully.
X-IBM-ClientSecret	Similar to a client password . Required to access the API	Y	Will be available after registering in the RMG API Developer portal.	The key pair created with registering the App on RM developer portal provides the client ID and Secret. The app needs to have a valid subscription to Order API in order to access the API successfully.
X-RMG-Date-Time	This should be populated with the date time in ISO 8601 format	Y	2022-05-10T10:04:00Z	
X-RMG-Language	Default english	N	en	
Accept	Content-Types that are acceptable for the response	N	application/json	
Body				

timeslotReservationId	Reservation ID obtained from the availability check API	Y	2020-09-19	There is a key dependency on Royal Mail Parcel Collect Order Handler API's /slots method before populating the date in this field. A "time slot reservation token" must be obtained by calling then Order Handler API's /slots method. The token obtained by calling the /slots method should be populated in this field and used within 15 minutes since the token expires after that. The token can only be used for one successful collection booking and a new one has to be obtained from /slots method for subsequent collection booking API calls.
collectionDate	The date on which the customer wants to item to be collected.	Y	2020-09-19	There is a key dependency on Royal Mail Parcel Collect Order Handler API's /slots method before populating the date in this field. A "time slot reservation token" must be obtained by calling then Order Handler API's /slots method and the date entered in this field MUST be one of the dates returned by the slots API. Otherwise this API request will fail.
suppressRmNotifications	This is a flag to indicate whether the consumer of this API want to use Royal Mail provisioned email notifications or not. Setting this field to "True" will suppress the Royal Mail triggered notifications and hence no emails	Y	"true" or "false"	Royal Mail triggered notification emails will have Royal Mail branding and follow a standard template.

	will be sent out.			
senderDetails		Y		
senderName	Name of the customer requesting for collection	Y	E.g. Mr. Pitney Jones	
senderEmail	Email Id of the customer requesting for collection	Y	E.g. pitneyjones@post.co.uk	This is the email address to which all email notifications will be sent - provided "suppressRmNotifications" field above has been set to False - E.g. Successful collection booking, item successfully collected, collection attempted, collection failed, etc.
accountDetails	Any one of the two fields below (retailerAccountNumber or partnerId) MUST be provided in the request. Otherwise the API request will fail and report an error.	Y		
retailerAccountNumber	Account number provided to the retailer when onboarded into RMG	Y (conditional)	E.g. 368322000	Only one of the two fields (either account number or partner ID) must be provided – not both fields. Partner ID should be populated only in case of postage bought via OLP channel.

partnerId	partnerId provided to the retailer when onboarded into RMG	Y (conditional)		Only one of the two fields (either account number or partner ID) must be provided – not both fields. Partner ID should be populated only in case of postage bought via OLP channel.
address	Populate the collection address in this section. The address provided here is the place from where the item(s) need to be collected and not necessarily the "sender address" specified when buying the postage/populating the labelInfo section below.	Y		If a valid postcode is not present in the address (either as part of postcode field or within one of the addressLine fields) the chances of collection request being rejected will be high as the address matching logic may not be able to output a single matching address.
addressLine1	Address Line 1 of the postal address provided by the customer	Y	E.g. Flat 45 Bellow House	Please populate valid data in this field in as many cases as possible for higher probability of acceptance of order.

	for collection			
addressLine2	Address Line 2 of the postal address provided by the customer for collection	N	E.g. Gayton Road	Please populate valid data in this field in as many cases as possible for higher probability of acceptance of order.
addressLine3	Address Line 3 of the postal address provided by the customer for collection	N		Please populate valid data in this field in as many cases as possible for higher probability of acceptance of order.
postTown	Postal town of the address provided by the customer for collection	Y	E.g. Harrow	Please populate valid data in this field in as many cases as possible for higher probability of acceptance of order.
county	Postal town of the address provided by the customer for collection	N	E.g. Middlesex	Please populate valid data in this field in as many cases as possible for higher probability of acceptance of order.
postcode	Post code of the provided by the customer for collection	Y	E.g HA1 2DQ	Please populate valid data in this field in as many cases as possible for higher probability of acceptance of order.

DPS	Two bit Delivery Point Suffix (DPS) for the collection address as per Royal Mail PAF.	Y	E.g 1A	The DPS must be obtained by the consumer of the API from either the Royal Mail PAF files or from an external address matching service provider who uses RM PAF as the source of addresses and DPS. This is essential since the combination of postcode and DPS will be used to determine the Royal Mail Delivery Office and delivery/collection routes prior to accepting the collection request.
safePlaceDetails		N		
locationText	Details of the safeplace collection	N	In the garage	
locationCode	For future use - can be ignored.	N		This field can be omitted from the request as it is intended for future use.
animalHazardDetails	Details of animals/pets at collection location	N	E.g. Two dogs at the property but kept restrained.	
items	Details of the parcel(s) that need to be collected from the customer	Y		This section is an array and hence can repeat if there are multiple items to be collected on the same collection date.
itemBarcode	1D Tracking ID of the RMG Returns Label (1D barcode number)	Y	WJ368776691GB	1D barcode number should be of length 13. First and the last two bits are characters and the others are numbers from the barcode range allocated to the retailer, specific to the RM product being used.
weightInGrams	Weight of the item in gms	Y		Max allowed weight is 20KG regardless of the postage size/weight category, like large letter, small parcel, medium parcel, etc. (the primary use of this field is only to determine whether an item will be too big/heavy to collect)

			E.g. 1570																
itemServiceName	Type of service	Y	E.g. Tracked Returns 48 (T48) Enhanced	This is the Royal Mail product family name of the postage. Currently allowed values for this field are: <table><tr><td>Royal Mail Tracked 48</td></tr><tr><td>Royal Mail Tracked 24</td></tr><tr><td>Royal Mail Tracked 48 with Signature</td></tr><tr><td>Royal Mail Tracked 24 with Signature</td></tr><tr><td>Royal Mail Special Delivery Guaranteed by 1 PM</td></tr><tr><td>Royal Mail Special Delivery Guaranteed by 9 AM</td></tr><tr><td>Royal Mail Special Delivery Guaranteed Returns</td></tr><tr><td>Royal Mail Standard</td></tr><tr><td>Royal Mail Signed for</td></tr><tr><td>International Signed</td></tr><tr><td>International Tracked</td></tr><tr><td>International Standard</td></tr><tr><td>International Import</td></tr><tr><td>Tracked Returns 24 (T24) Enhanced</td></tr><tr><td>Tracked Returns 48 (T48) Enhanced</td></tr></table>	Royal Mail Tracked 48	Royal Mail Tracked 24	Royal Mail Tracked 48 with Signature	Royal Mail Tracked 24 with Signature	Royal Mail Special Delivery Guaranteed by 1 PM	Royal Mail Special Delivery Guaranteed by 9 AM	Royal Mail Special Delivery Guaranteed Returns	Royal Mail Standard	Royal Mail Signed for	International Signed	International Tracked	International Standard	International Import	Tracked Returns 24 (T24) Enhanced	Tracked Returns 48 (T48) Enhanced
Royal Mail Tracked 48																			
Royal Mail Tracked 24																			
Royal Mail Tracked 48 with Signature																			
Royal Mail Tracked 24 with Signature																			
Royal Mail Special Delivery Guaranteed by 1 PM																			
Royal Mail Special Delivery Guaranteed by 9 AM																			
Royal Mail Special Delivery Guaranteed Returns																			
Royal Mail Standard																			
Royal Mail Signed for																			
International Signed																			
International Tracked																			
International Standard																			
International Import																			
Tracked Returns 24 (T24) Enhanced																			
Tracked Returns 48 (T48) Enhanced																			
itemStatus	Initial status of the booking request	Y	AwaitingCollection	Populate the field with the string "AwaitingCollection"; Do not use any other value/text for this field.															
itemProductCode	retailer specific ID e.g. Order number / invoice number	N		Provisional field for cross-referencing the collection booking request against a retailer/3PI specific reference number - intended for future use.															
itemType	Type of collection for the item	N	E.g NoLabel	(Enumerated field) Populate with NoLabel for label-less collections; Standard for print-at home collections;															

dimension s	Dimensio ns of the parcel(in cms)	Y		
height	Height/Le ngth of the parcel (Max 61cm)	Y	E.g. 30	Max allowed height/length is 61cm regardless of the postage size/weight category, like large letter, small parcel, medium parcel, etc. (the primary use of this field is only to determine whether an item will be too big/heavy to collect)
width	Width of the parcel (max 46cm)	Y	E.g. 20	Max allowed width is 46 cm regardless of the postage size/weight category, like large letter, small parcel, medium parcel, etc. (the primary use of this field is only to determine whether an item will be too big/heavy to collect)
depth	Depth of the parcel (max 46cm)	Y	E.g. 12	Max allowed depth is 46cm regardless of the postage size/weight category, like large letter, small parcel, medium parcel, etc. (the primary use of this field is only to determine whether an item will be too big/heavy to collect)
labelInfo		N		This section must be populated for label- less collections. It should not be populated for Standard (print-at-home) collections. The contents of this section are purely for the purpose of generating a printable label file on Royal Mail side, so that the postie can bring the label to point of collection. Some fields might be same as the fields populated in the sections above.
1DTracking Number	1d tracking ID (same as the "itemBarc odeld" in the section above)	Y		13 digit 1D barcode number - POPULATE THE SAME VALUE AS "itemBarcodeId".
2DUniquel dentifier	2D tracking ID - 21 digit 2D barcode id that appears above the QR code	Y		This is not a field for populating the binary image of the QR code, but for the 21 digit 2D barcode that appears above the QR code on the label.

	on the label;			
postByDate	Post by date applicable to the label	Y	E.g. 2022-05-17	Typically two to three weeks from the date the collection is being booked on.
RMService	Type of postage - same as itemServiceName in the section above.	Y	Tracked Returns 48 (T48) Enhanced	Same as itemServiceName field in the section above. For tracked returns 48 items, use the exact same string specified in the sample value column.
pricePaid	Price paid by the customer for the postage	Y	E.g 0	Populate this field with 0 for retailer paid Returns items. But for OLP / outbound items the actual price paid for booking a doorstep collection should be entered here (if any).
referenceNumber	Retailer specific reference number that needs to be printed on the label	N		E.g. Order number, invoice number, returns reference number, etc.
displayReferenceAsBarcode	If the reference number needs to be displayed as barcode	N	FALSE	Set this to false
referenceText	Retailer specific text that needs to be printed on the label	N		
weightInGrams	Weight of the item in gms	Y	E.g. 1570	

itemForma t	package size/form at based on dimensio ns	Y	- Letter - LargeLetter - SmallParcel - MediumPa rcel	Enumerated field - use one the four types as relevant
Recipient Address	Standard UK address	Y		UK address comprising of fields : <pre> { "companyName": "string", -optional "addressLine1": "string", -required "addressLine2": "string", -optional "addressLine3": "string", -optional "postTown": "string", -required "county": "string", -optional "postcode": "string", -required "countryCode": "GBR" }</pre>
Sender Address	Standard UK address	Y		

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